FinTech Lab 1
Report

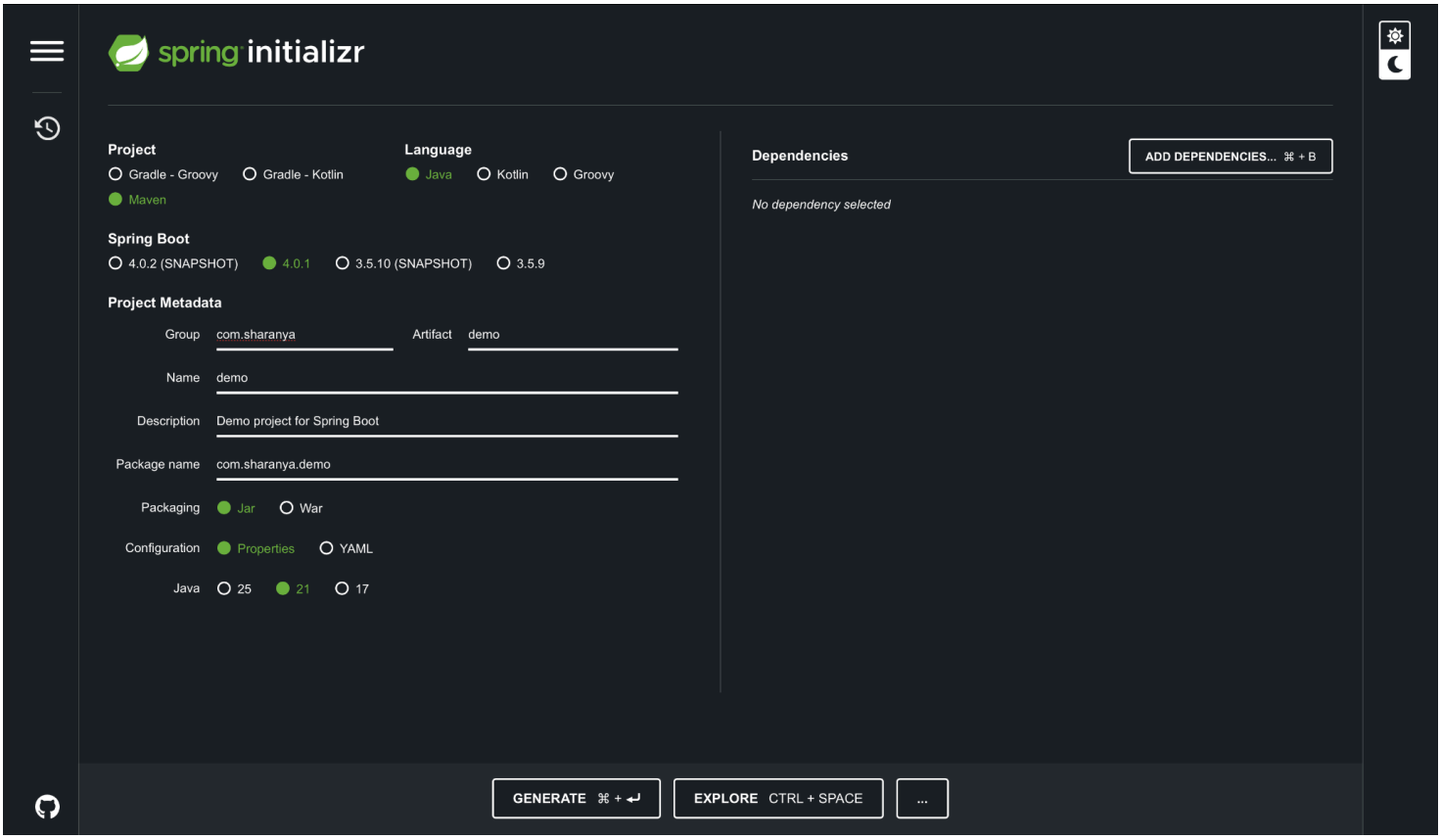
Objective:

To set up a Spring Boot application using Spring Initializr, display a Hello World message, organize the logic in a separate controller package, and test the application using Postman.

Steps Followed:

Step 1: Creation of Spring Boot Project

The Spring Boot project was created using Spring Initializr by visiting https://start.spring.io/. The project type was selected as Maven with Java as the programming language. The latest stable version of Spring Boot was chosen. The group name, artifact name, and package name were provided appropriately. Jar packaging was chosen along with Java 17. The Spring Web dependency was added. The project was then generated and extracted.

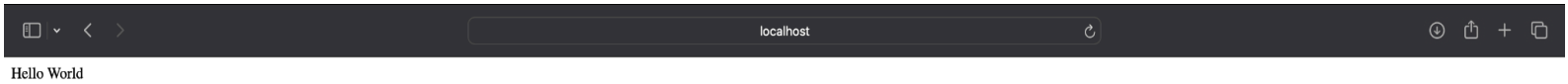


Step 2: Importing the Project into IDE

The extracted project folder was opened in an IDE, in this case, VS Code. Maven automatically downloaded all required dependencies and the project structure was successfully loaded.

Step 3: Running the Spring Boot Application

The main application file located inside the **src/main/java** directory was executed. The embedded Tomcat server started successfully on port **8080**, indicating that the application was running correctly.



Hello World

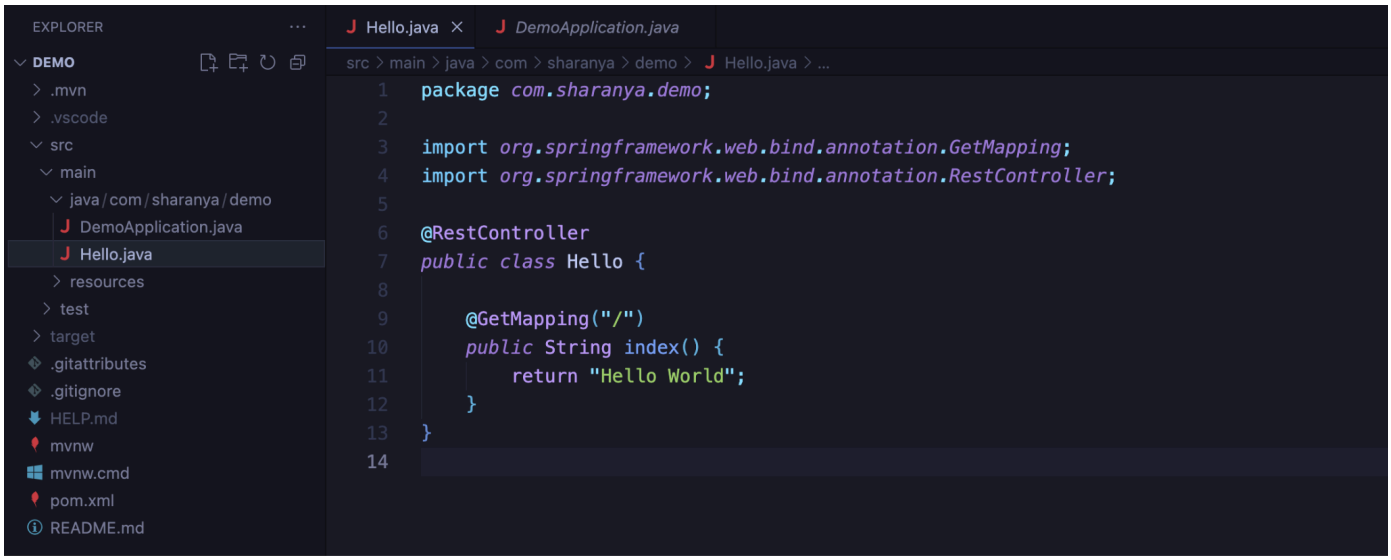
Procedure

1. Set up Spring boot application

The Spring Boot application was set up using Spring Initializr and executed. Successful startup of the application confirmed proper configuration of the project.

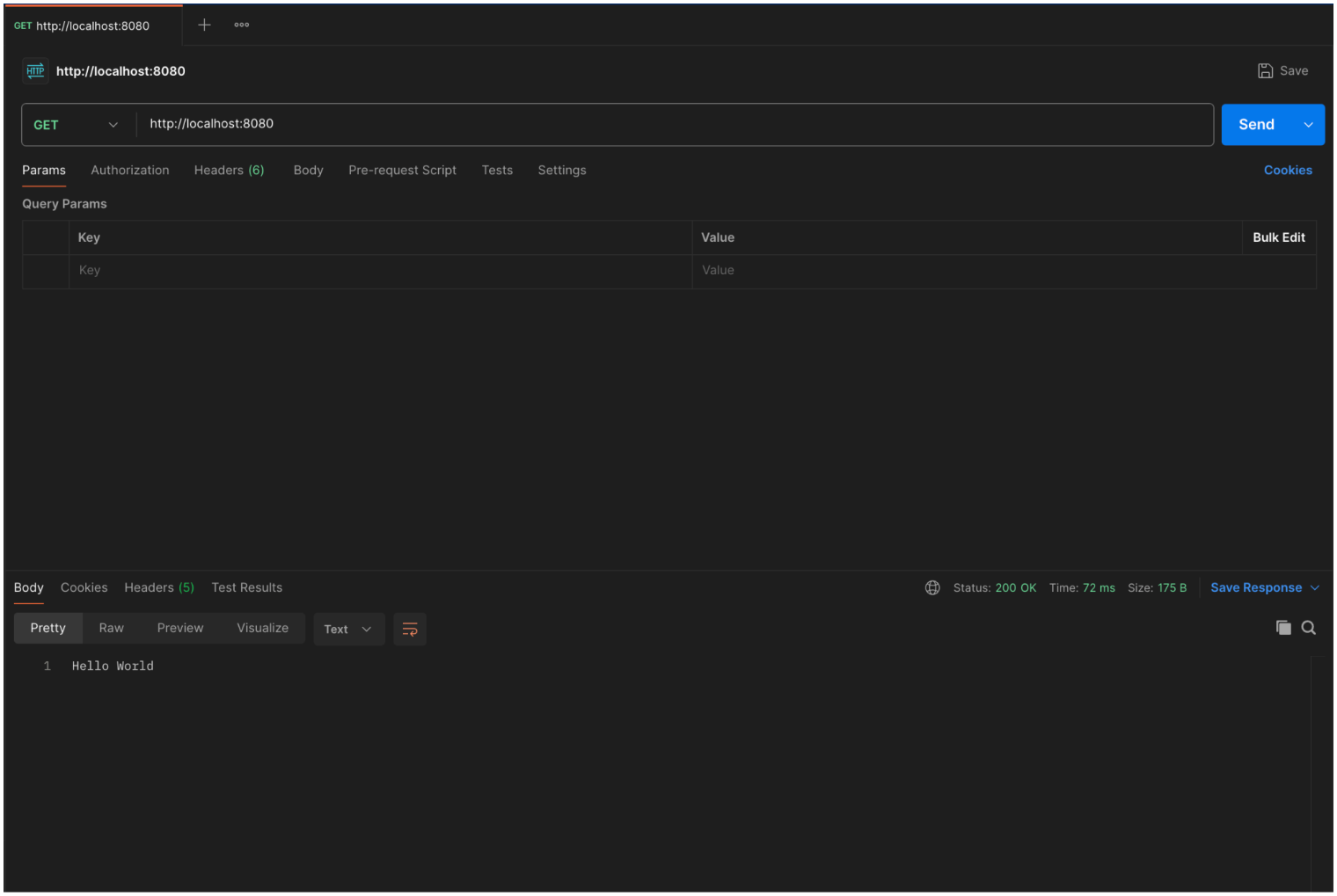
2. Create a Controller package

A separate package named **controller** was created within the base package. A REST controller class was added inside this package. The class was annotated with `@RestController` and a method was mapped using @GetMapping. The method returned the message **"Hello World"** as the response.



3. Application testing using Postman

The application, hosted locally, was tested using Postman. A GET request was sent to the endpoint `http://localhost:8080/hello`. The response was received successfully, confirming that the controller and thus the endpoint were functioning correctly.



Lab Exercises – Answers

1. Setup the Spring Boot application and print the Hello World message

The Spring Boot application was successfully created and executed. The hello world message was displayed, confirming correct application setup.

2. Setup the Spring Boot application with the hello world message in a separate package named controller

The application logic was organized by creating a separate controller package. The hello world message was returned through a REST controller, ensuring clean separation of concerns.

3. Test the above application using Postman

The application was tested using Postman by sending a GET request to the configured endpoint. The correct response was received, verifying successful implementation.

Output-

The Spring Boot application was successfully created, executed, and tested. The hello world message was displayed through a REST API and verified using Postman.